

Medical Adherence App - Project Overview

2-Page High SNR Reference | Version 1.0 | Android App

What It Is

Clean, minimal Android medication tracking app designed for **elderly users**. Built with Jetpack Compose, Material 3, and Room database. Features large touch targets ($\geq 48dp$), adjustable fonts, and non-judgmental language.

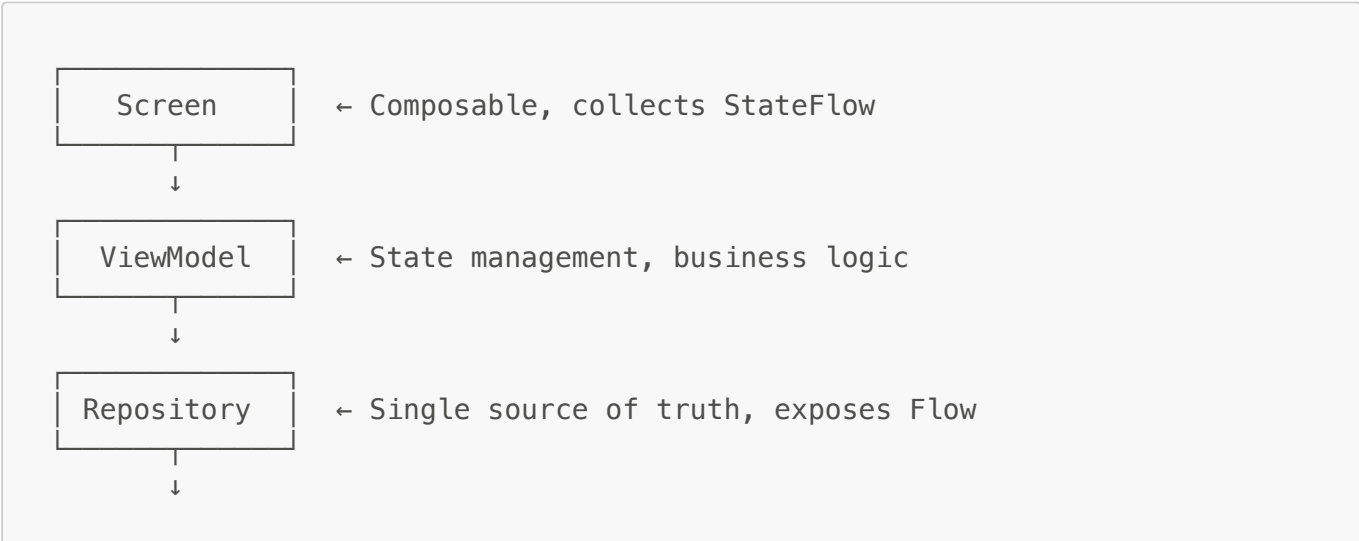
Core Function: Track daily medications → Monitor adherence with stats → Build consistency with streaks

Tech Stack (Essential)

Category	Technology	Version
Language	Kotlin	2.0.21
UI	Jetpack Compose + Material 3	BOM 2024.09.00
Architecture	MVVM (ViewModel + StateFlow)	Lifecycle 2.9.4
Database	Room (SQLite)	2.6.1
Navigation	Navigation Compose	2.8.5
Async	Kotlin Coroutines + Flow	1.9.0
Min SDK	Android 10 (API 29)	-
Target SDK	Android 15+ (API 36)	-

Build: Gradle 8.13.0 with KSP for Room annotation processing

Architecture (MVVM Pattern)



Room (DAO)

← Database queries, persistence

Data Flow: User Action → ViewModel → Repository → Room → Flow → StateFlow → Screen Recompose

Key Files:

- **MainActivity.kt** - Single activity + NavHost
- **HomeScreen.kt** - Daily dose tracking UI
- **HomeViewModel.kt** - State management with StateFlow
- **MedicationRepository.kt** - Data operations
- **AppDatabase.kt** - Room database definition

Features (User Perspective)

1. Home Screen (Daily Tracking)

- **Live countdown** to next dose
- **Dose cards** with 3 action buttons: Taken / Missed / Snooze 15m
- **Weekly stats** at top: Adherence % + Streak days
- **Undo button** for corrections
- **FAB (+)** to add medications

2. Medications Library

- View all medications with full schedules
- Edit/delete via overflow menu (:)
- Support for multiple doses per day
- Frequency options: Daily, Specific Days, Weekly, Every X Days, As Needed

3. Statistics

- **Weekly percentage:** 0-100% adherence with color-coded progress bar
- **Streak counter:** Consecutive days of 100% adherence
- **Daily bar chart:** Visual breakdown for Mon-Sun
- **Encouraging feedback:** Based on performance (non-judgmental)

4. Settings

- **Font size:** Normal (1.0x) or Large (1.15x)
- **Theme:** Follows system light/dark mode

Data Models

Medication

```
data class Medication(  
    val id: String,  
    val name: String,           // "Amlodipine"  
    val dosage: String,        // "5 mg"  
    val times: List<String>,    // ["07:00", "19:00"]  
    val notes: String?,        // "Take with food"  
    val frequency: MedicationFrequency,  
    val specificDays: List<Int> // [1,3,5] for Mon/Wed/Fri  
)
```

DoseEvent

```
data class DoseEvent(  
    val medId: String,  
    val date: LocalDate,  
    val time: String,           // "07:00"  
    val taken: Boolean           // true=taken, false=missed  
)
```

Room Entities: `MedicationEntity`, `DoseEventEntity` (with type converters for lists)

Project Structure

```
app/src/main/java/com/example/medicaladherence/  
├── MainActivity.kt           # Single activity + navigation  
├── data/  
│   ├── model/               # Medication, DoseEvent, enums  
│   ├── local/  
│   │   ├── AppDatabase.kt   # Room database  
│   │   ├── Converters.kt    # Type converters  
│   │   ├── entity/          # Room entities  
│   │   └── dao/              # MedicationDao, DoseEventDao  
│   └── repository/  
│       └── MedicationRepository.kt # Single source of truth  
├── viewmodel/               # HomeViewModel, StatsViewModel,  
etc.  
└── ui/  
    ├── screens/              # HomeScreen, StatsScreen, etc.  
    ├── components/           # DoseCard (reusable)  
    └── theme/                 # Material 3 theme
```

Key Technical Patterns

StateFlow for Reactive UI

```
// ViewModel
private val _uiState = MutableStateFlow(HomeUiState())
val uiState: StateFlow<HomeUiState> = _uiState.asStateFlow()

// Screen
val uiState by viewModel.uiState.collectAsState()
```

Room with Flow

```
@Dao
interface MedicationDao {
    @Query("SELECT * FROM medications")
    fun getAllMedications(): Flow<List<MedicationEntity>>
}
```

Repository Pattern

```
class MedicationRepository(database: AppDatabase) {
    val medications: Flow<List<Medication>> =
        medicationDao.getAllMedications().map { it.toMedication() }
}
```

Accessibility Features

Feature	Implementation
Large Touch Targets	All buttons ≥48dp
Font Scaling	Settings: Normal (1.0x) or Large (1.15x)
Color Contrast	WCAG AA compliant, Material 3
Calm Palette	Soothing blue, no alarming colors
Friendly Language	"Missed" not "Failed", encouraging feedback
TalkBack	Screen reader compatible

Quick Start

For Users

- 1. Tap **+** **button** → Add medication (name, dosage, times)
- 2. Home screen shows today's doses
- 3. Tap **Taken** when you take medication

4. Check **Stats** tab for weekly progress

For Developers

```
# Clone and open in Android Studio
cd Medical_Adherence
./gradlew build

# Run on device/emulator
./gradlew installDebug

# Run tests
./gradlew test
```

Requirements: Android Studio Hedgehog+, SDK 29+, Kotlin 2.0.21

Routes & Navigation

HOME = "home"	# Daily tracking
MEDICATIONS = "medications"	# Library view
ADD_MEDICATION = "add_medication"	# Add/edit form
STATS = "stats"	# Weekly statistics
SETTINGS = "settings"	# Font & theme

Bottom Nav: Home |  Medications |  Stats |  Settings

Database Schema

Tables

- **medications**: id, name, dosage, times, notes, frequency, specificDays
- **dose_events**: id, medId, date, time, taken, timestamp
- **settings**: id, fontScale

Type Converters

- **List<String>** → JSON string
 - **List<Int>** → comma-separated
 - **MedicationFrequency** → String
 - **LocalDate** → Long (epoch day)
-

State Management

UI State Classes

```
data class HomeUiState(  
    val todayDoses: List<DoseItem>,  
    val nextDoseCountdown: String,    // "3h 24m"  
    val weeklyAdherencePercent: Int,  // 0-100  
    val streakDays: Int,  
    val snackbarMessage: String?  
)
```

Lifecycle

- ViewModel survives configuration changes
- Room queries automatically stop when screen not visible
- Coroutines cancelled when ViewModel cleared

Deployment Info

Package: `com.example.medicaladherence` **Version Code:** 1 **Version Name:** "1.0"

Build Variants:

- Debug: Development with full logging
- Release: Production (minification currently disabled)

No external dependencies: Pure Jetpack stack, local storage only

Future Enhancements

- Push notifications for dose reminders (WorkManager)
- Data export to CSV/PDF
- Multi-user support with profiles
- Medication refill tracking
- Health data API integration

Key Metrics

- **5 screens:** Home, Medications, Stats, Settings, Add/Edit
- **3 tables:** medications, dose_events, settings
- **5 ViewModels:** One per main screen
- **2 DAOs:** MedicationDao, DoseEventDao
- **1 Repository:** MedicationRepository (single source of truth)

Documentation: See [docs/user/](#) for user guides, [docs/technical/](#) for deep dives

License: Prototype for demonstration purposes